Object-Oriented Programming and Design



Software College of NEU



Now, you should be able to



- Understand the logic behind program design
- Know the core programming concepts and the basic process for developing a program
- Establish effective programming skills to solve simple problems
- Learn a new programming language by yourself

Objectives of this course

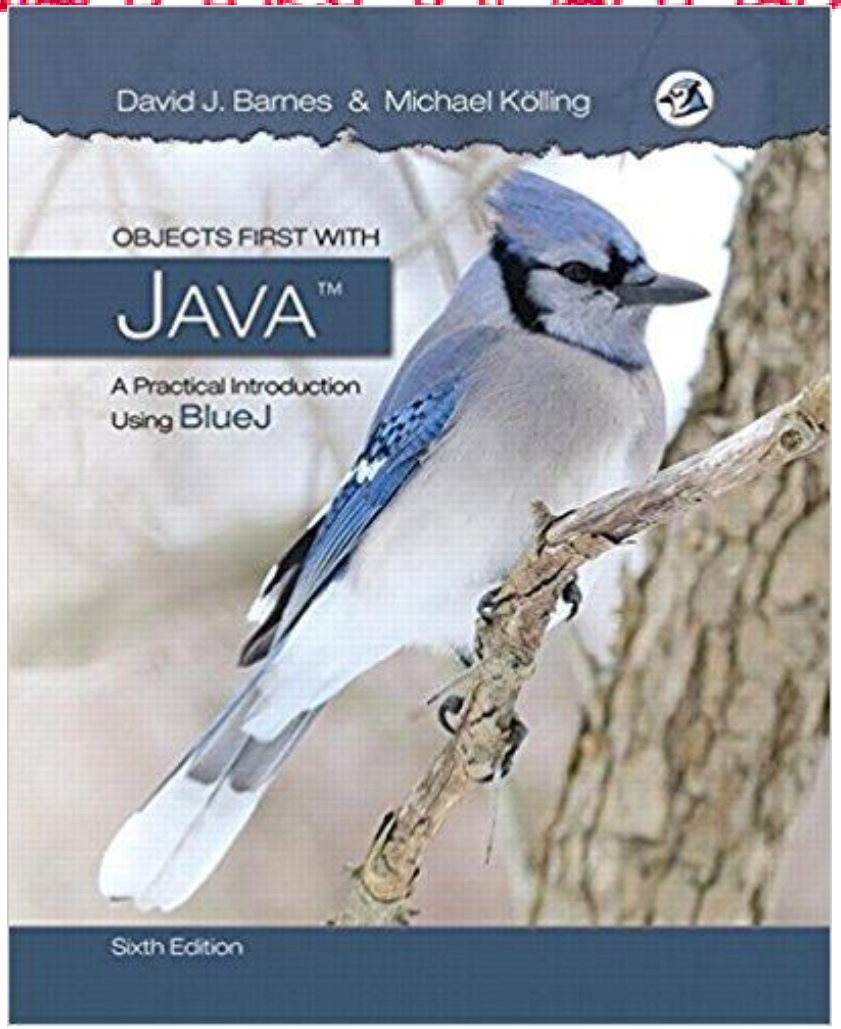


- To provide an understanding of the essentials of object oriented programming.
- To provide an understanding of the fundamentals of object oriented design.
- Use *Java* to create object oriented applications.

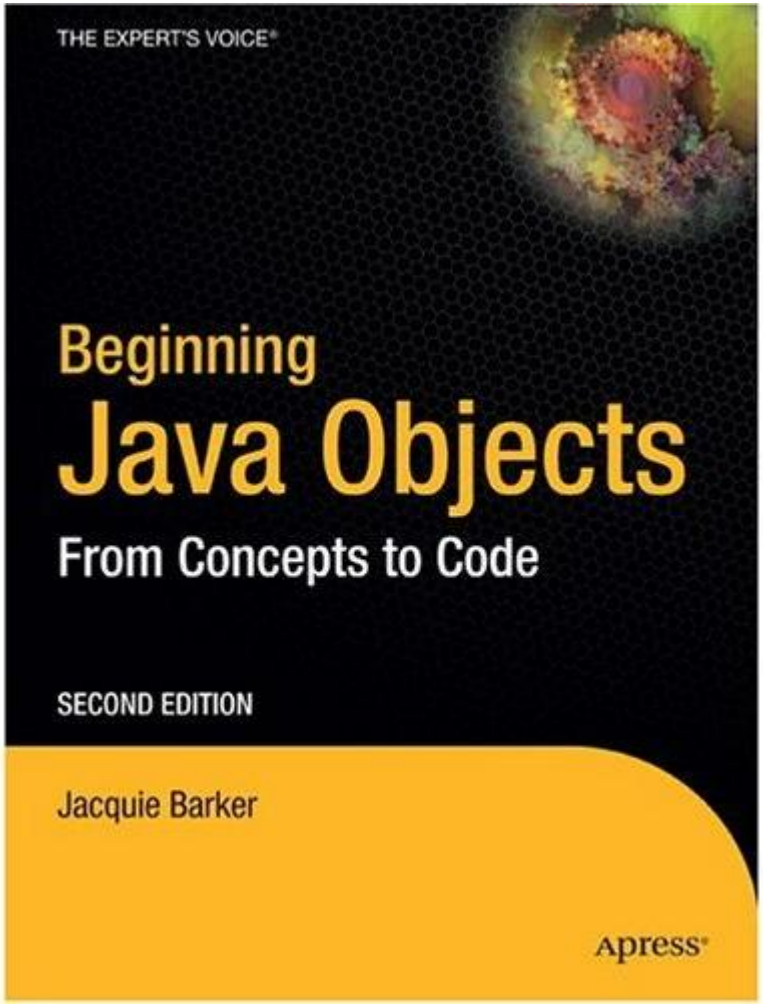
After learning this course, you should be able to solve a more complex engineering problem.

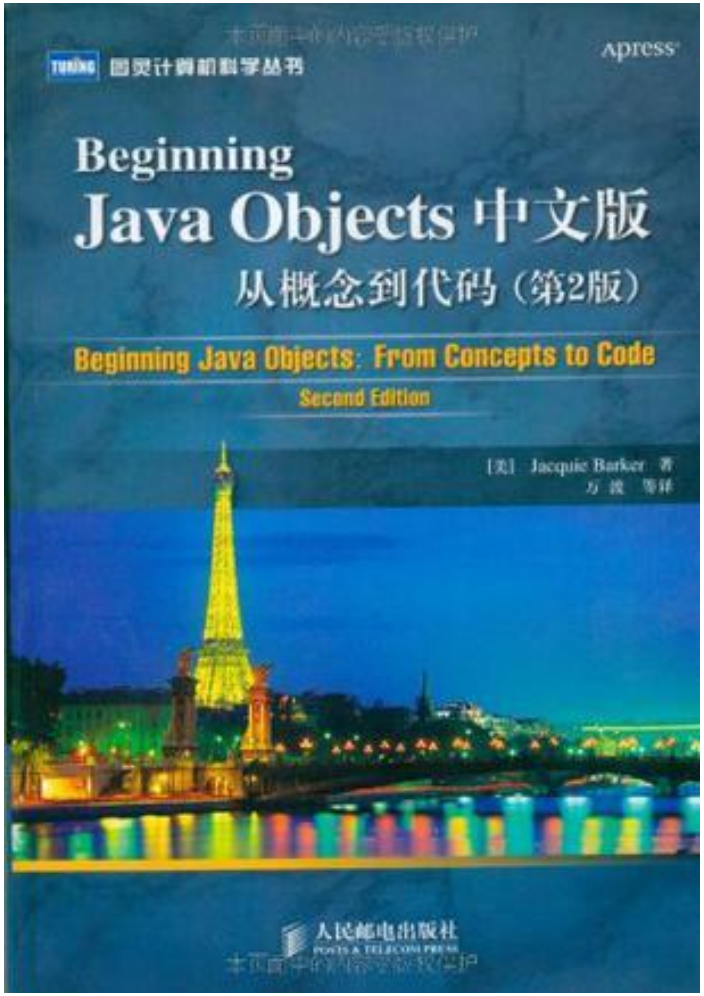
Textbook

- **Barnes, David J. Objects First with Java: A Practical Introduction Using BlueJ. Sixth Edition. Pearson, 2016.**



References


THE EXPERT'S VOICE®
Beginning
Java Objects
From Concepts to Code
SECOND EDITION
Jacquie Barker
Apress®


TURING 图灵计算机科学丛书
Apress®
Beginning
Java Objects 中文版
从概念到代码（第2版）
Beginning Java Objects: From Concepts to Code
Second Edition
[美] Jacquie Barker 著
人民邮电出版社

References

- Java official website
  - https://www.oracle.com/technetwork/java/index.html
- 中国大学MOOC(https://www.icourse163.org)
  - Java basics: https://www.icourse163.org/course/ZJU-1001541001
  - Object-oriention (Java): https://www.icourse163.org/course/ZJU-1001542001
  - Java programing: https://www.icourse163.org/course/PKU-1001941004
  - Object-oriented analysis and design: https://www.icourse163.org/course/ECNU-1003434002
- BB平台
  - http://bb.neu.edu.cn

Evaluation

- **Experiments - 20%**
- **Homework - 20%**
- **Examination - 60%**

Activity	Total Hours
Lectures	40 lectures
Labs	8 labs

- **You are expected to attend each class. If you are unable to attend, please let me know.**

Outline



- Ch 0: Introduction (2h)
- Ch 1: Java Language Overview (4h)
- Ch 2: Objects and Classes (6h)
- Ch 3: Object Interactions (4h)
- Ch 4: Relationships Between Objects (8h)
- Ch 5: Designing Classes (6h)
- Ch 6: Advanced Class Design (4h)
- Ch 7: Exception (2h)
- Ch 8: File I/O (4h)

A short review by learning Java basics

Programming concepts in Object-Orientation

Design principles in Object-Orientation

Programming skills in Java

Learning outcomes





At the end of the unit students should be able to:

1. **Explain the principles of the object-oriented programming paradigm specifically including abstraction, encapsulation, inheritance and polymorphism, and explain how these principles are used to create object oriented programs.**
2. **Design, develop, test, and debug programs using object-oriented principles in conjunction with an integrated development environment.**
3. **Select and use appropriate collection classes, from the languages class library, to manage collections of multiple objects.**
4. **Construct appropriate diagrams and textual descriptions to communicate the static structure and dynamic behavior of an object-oriented solution.**
5. **Apply accepted good practices related to the construction of object-oriented programs.**

How should you learn this course?



- You are encouraged to learn more by yourself
- Please be prepared well before each class by getting more information from the Web and answering the assigned questions
- Please take an active part in the classroom discussion
- Please always make a conclusion before the end of each class
- You are always welcome to ask question

Question?



Programming

is a

practical skill

that is

mastered

by

doing it,

not just

reading about it!